RQ2_Brian

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Libraries

```r
pkg_needed <- c("tidyverse", "cluster", "factoextra", "pheatmap", "ggrepel")
for (p in pkg_needed) {
  if (!requireNamespace(p, quietly = TRUE)) {
    install.packages(p)
  }
}

library(pheatmap)
```

```
## Warning: package 'pheatmap' was built under R version 4.5.3
```

```r
library(tidyverse)
```

```
## -- Attaching core tidyverse packages ------------------------ tidyverse 2.0.0 --
## v dplyr     1.2.0     v readr     2.1.6
## v forcats   1.0.1     v stringr   1.6.0
## v ggplot2   4.0.1     v tibble    3.3.1
## v lubridate 1.9.5     v tidyr     1.3.2
## v purrr     1.2.1
## -- Conflicts ------------------------------------------ tidyverse_conflicts() --
## x dplyr::filter() masks stats::filter()
## x dplyr::lag()    masks stats::lag()
## i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all conflicts to become errors
```

```r
library(cluster)
```

```
## Warning: package 'cluster' was built under R version 4.5.3
```

```r
library(factoextra)
```

```
## Warning: package 'factoextra' was built under R version 4.5.3

## Welcome to factoextra!
## Want to learn more? See two factoextra-related books at https://www.datanovia.com/en/product/practica
```

```r
library(here)
```

```
## here() starts at C:/Users/win4k/Documents/glassdoor-company-reviews
```

```r
library(cluster)
library(ggrepel)
```

```
## Warning: package 'ggrepel' was built under R version 4.5.3
```

For Cleaner Plots

```r
theme_set(
  theme_minimal(base_size = 13) +
    theme(
      plot.title = element_text(face = "bold", size = 15),
      plot.subtitle = element_text(size = 11),
      axis.title = element_text(face = "bold"),
      panel.grid.minor = element_blank(),
      legend.position = "bottom"
    )
)
```

Read from current data version

```r
df <- read.csv(here("data", "processed", "glassdoor_reviews_cleaned_v2.csv"))

dim(df)
```

```
## [1] 1000   32
```

```r
head(df)
```

```
##   review_id rating_date count_helpful count_unhelpful employee_job_end_year
## 1  28951745  2019-08-30            10               0                  2019
## 2  59978917  2022-02-19             2               0                    NA
## 3  42749601  2021-02-19             0               0                  2021
## 4  12095563  2016-09-27             4               0                  2015
## 5  46084540  2021-04-26             0               0                    NA
## 6  58316203  2022-01-21             1               0                  2021
##   employee_length employee_status covid_related_review flags_business_outlook
## 1              20         REGULAR                    0               NEGATIVE
## 2               6         REGULAR                    0               POSITIVE
## 3               2         REGULAR                    1               POSITIVE
## 4               2         REGULAR                    0               NEGATIVE
## 5               2         REGULAR                    1               NEGATIVE
## 6               2                                    0               POSITIVE
```

```
##   flags_ceo_approval flags_recommend_frend rating_culture_values
## 1         DISAPPROVE              NEGATIVE                     1
## 2            APPROVE              POSITIVE                     3
## 3            APPROVE              POSITIVE                     5
## 4         NO_OPINION              NEGATIVE                     1
## 5            APPROVE              NEGATIVE                     5
## 6            APPROVE              POSITIVE                     5
##   rating_diversity_inclusion rating_overall rating_work_life
## 1                          3              2                1
## 2                          5              5                2
## 3                          5              4                5
## 4                          5              2                1
## 5                          5              3                3
## 6                          5              5                5
##                                                                    summary
## 1                                                       IT is Out of Control
## 2                                                  Great Company to work with
## 3                                                                      Great
## 4 Vodafone is an Egotistical Company that only cares about its net promoter score
## 5                                 Great company, poor processes, high expectations
## 6                                                   Commercial Product Internship
##      company_name   employee_location       employee_job_title
## 1 Elevance Health                                     Manager
## 2       HDFC Bank           New Delhi Assistant Vice President
## 3          Google             Chennai       Software Developer
## 4        Vodafone     London, England     Retail Store Manager
## 5         Verizon Rolling Meadows, IL Business Account Manager
## 6    Florida Blue
##
## 1                                Humble yourselves. Remember when you were coming up in your career. Re
## 2
## 3
## 4  You have some amazing staff members who are totally disengaged. Your TNPS system is flawed. Your c
## 5 Handle customer escalations so BAMs can focus on sales. Be realistic about sales goals, especially
## 6
##
## 1                                             Salary was good for the position, but only because they
## 2                Great brand name. Top Management are consistent on bank's goals and strategies. Cons
## 3                                                                                          Good an
## 4                                                   Unfortunately there were no pros during my time wo
## 5 Working with business owners or employees to solve telecommunication and technology. Great salary,
## 6                                  Very inclusive and helpful environment which made succeeding in th
##
## 1
## 2
## 3
## 4                                         Retail stores are severely understaffed. The new system, So
## 5 Sales expectations are unrealistic. Due to Covid, the job shifted from outside sales to inside sale
## 6
##   rating_compensation_benefits rating_senior_leadership
## 1                            2                        1
## 2                            3                        5
## 3                            5                        5
## 4                            2                        2
```

```
## 5                            4                        2
## 6                            5                        5
##   rating_career_opportunities             sector rating_year helpfulness_score
## 1                           1         Healthcare        2019                10
## 2                           5            Finance        2022                 2
## 3                           3         Technology        2021                 0
## 4                           2 Telecommunications        2016                 4
## 5                           3 Telecommunications        2021                 0
## 6                           3            Finance        2022                 1
##   career_missing employee_job_quit employee_job_start_year has_review_text
## 1          FALSE              TRUE                    1999            TRUE
## 2          FALSE             FALSE                    2016            TRUE
## 3          FALSE              TRUE                    2019            TRUE
## 4          FALSE              TRUE                    2013            TRUE
## 5          FALSE             FALSE                    2019            TRUE
## 6           TRUE              TRUE                    2019            TRUE
```

1. Feature Engineering

Variables that matter for culture similarity

```r
# Select variables related to company culture / employee experience
glassdoor_model <- df %>%
  select(
    company_name,
    rating_work_life,
    rating_culture_values,
    rating_senior_leadership,
    rating_career_opportunities,
    rating_compensation_benefits,
    rating_overall,
    rating_diversity_inclusion,
    flags_recommend_frend,
    flags_business_outlook,
    flags_ceo_approval,
    employee_status
  )

# Check missing values
colSums(is.na(glassdoor_model))
```

```
##                 company_name             rating_work_life
##                            0                            0
##        rating_culture_values     rating_senior_leadership
##                            0                            0
##  rating_career_opportunities rating_compensation_benefits
##                            0                            0
##               rating_overall   rating_diversity_inclusion
##                            0                            0
##        flags_recommend_frend       flags_business_outlook
##                            0                            0
```

```
##           flags_ceo_approval              employee_status
##                            0                            0
```

Convert categorical sentiment variables into numeric form

```r
# Recode recommendation / outlook / CEO approval into numeric variables
glassdoor_model <- glassdoor_model %>%
  mutate(
    recommend_num = ifelse(flags_recommend_frend == "POSITIVE", 1, 0),

    outlook_num = case_when(
      flags_business_outlook == "POSITIVE" ~ 1,
      flags_business_outlook == "NEUTRAL"  ~ 0.5,
      flags_business_outlook == "NEGATIVE" ~ 0,
      TRUE ~ NA_real_
    ),

    ceo_num = case_when(
      flags_ceo_approval == "APPROVE"    ~ 1,
      flags_ceo_approval == "NO_OPINION" ~ 0.5,
      flags_ceo_approval == "DISAPPROVE" ~ 0,
      TRUE ~ NA_real_
    )
  )
```

Clustering methods need numeric inputs. This is reasonable because:

Positive sentiment = larger value Negative sentiment = smaller value Neutral or no opinion = middle value

Aggregate the data to the company level

```r
company_data <- glassdoor_model %>%
  group_by(company_name) %>%
  summarise(
    work_life = mean(rating_work_life, na.rm = TRUE),
    culture = mean(rating_culture_values, na.rm = TRUE),
    leadership = mean(rating_senior_leadership, na.rm = TRUE),
    career = mean(rating_career_opportunities, na.rm = TRUE),
    compensation = mean(rating_compensation_benefits, na.rm = TRUE),
    overall = mean(rating_overall, na.rm = TRUE),
    diversity = mean(rating_diversity_inclusion, na.rm = TRUE),
    recommend = mean(recommend_num, na.rm = TRUE),
    outlook = mean(outlook_num, na.rm = TRUE),
    ceo_approval = mean(ceo_num, na.rm = TRUE),
    review_count = n()
  )

nrow(company_data)
```

```
## [1] 693
```

The data is currently at the review level, meaning each row is one employee review.

But the question is about similar companies, so the clustering must happen at the company level.

This block creates one row per company, where each feature is the average across that company's reviews.

Remove companies with too few reviews

```
company_data_filtered <- company_data %>%
  filter(review_count >= 2)

# Check how many companies remain
nrow(company_data_filtered)
```

```
## [1] 72
```

A minimum review threshold was applied to ensure stability in company-level averages. While higher thresholds (e.g., 5 reviews) produce more reliable estimates, they significantly reduce the number of companies available for analysis. A threshold of at least 2 reviews was selected to balance reliability and sample size, resulting in 72 companies included in the clustering analysis. This choice allows for more meaningful cluster structure while acknowledging some variability in companies with fewer reviews.

Create the feature matrix for clustering

```
clustering_features <- company_data_filtered %>%
  select(
    work_life,
    culture,
    leadership,
    career,
    compensation,
    overall,
    diversity,
    recommend,
    outlook,
    ceo_approval
  )
```

Scale the Features

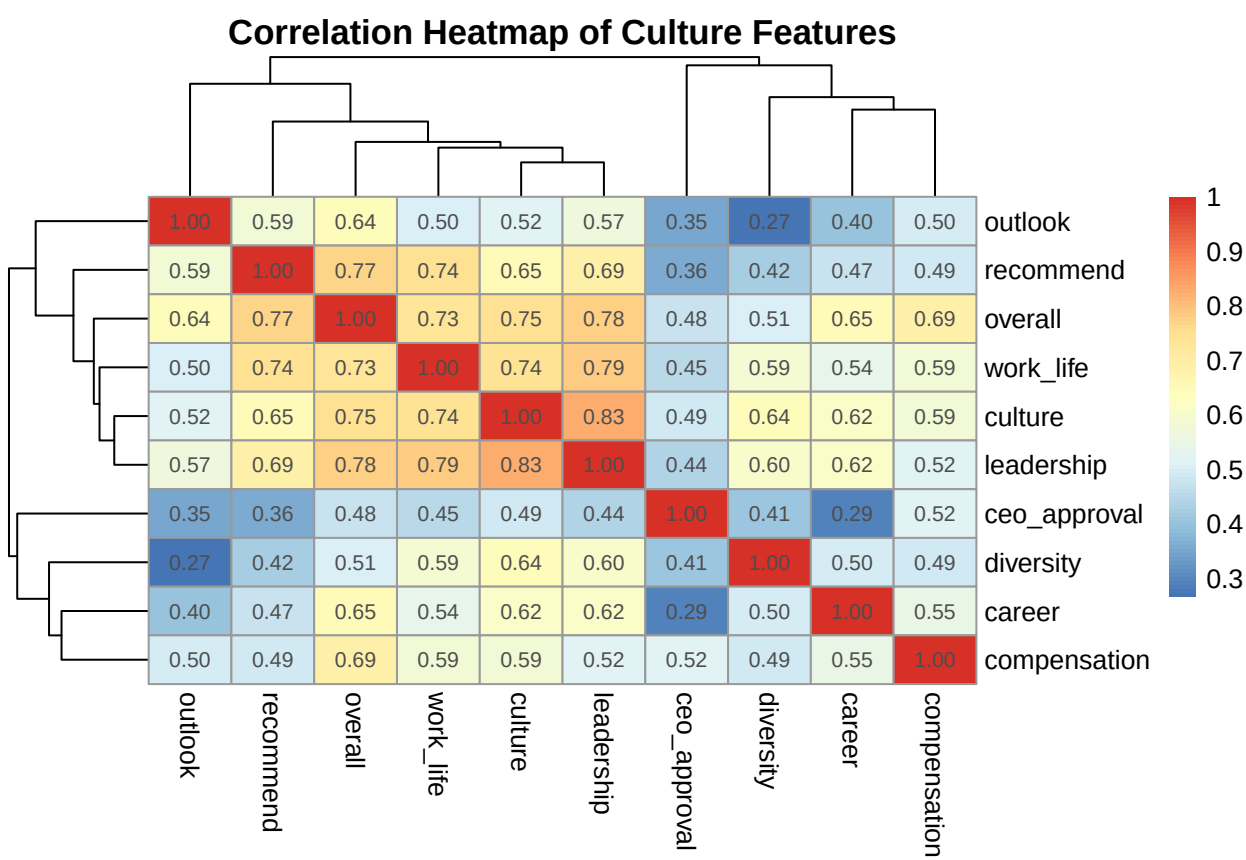
```
scaled_features <- scale(clustering_features)

# Look at first few rows
head(scaled_features)
```

```
##        work_life    culture leadership      career compensation     overall
## [1,] -0.05713474  0.5025266  0.4550375 -0.09327828  -0.26030561  0.06037578
## [2,]  0.40346468  0.9378494  0.8557675  0.87615114   0.16070899  0.69186031
```

```
## [3,]  1.96950269  1.2392268  1.5570451  0.66841626   2.05527467  1.79695823
## [4,]  0.19619494  0.4104391  1.0060413 -0.85497281  -0.54975314 -0.22905463
## [5,] -0.92569364 -0.5762128 -0.4108256 -0.41971879  -0.01220772 -0.10501302
## [6,]  1.96950269  2.3442770  2.1080488  0.66841626   0.89748453  1.79695823
##       diversity   recommend   outlook ceo_approval
## [1,] -0.1487798  0.64966268 0.5255959   -1.0967967
## [2,]  0.1469005  1.15804359 0.7616179    0.4541738
## [3,]  0.5017169  1.76810068 1.8237168    1.5279225
## [4,]  0.5017169  0.09044368 0.3633308    0.5436528
## [5,] -0.4740282 -0.86821746 0.4328730   -0.4406169
## [6,]  1.4774619  1.76810068 1.8237168    1.5279225
```

Feature Correlation Analysis

```r
feature_cor <- cor(clustering_features)

pheatmap(
  feature_cor,
  main = "Correlation Heatmap of Culture Features",
  display_numbers = TRUE
)
```



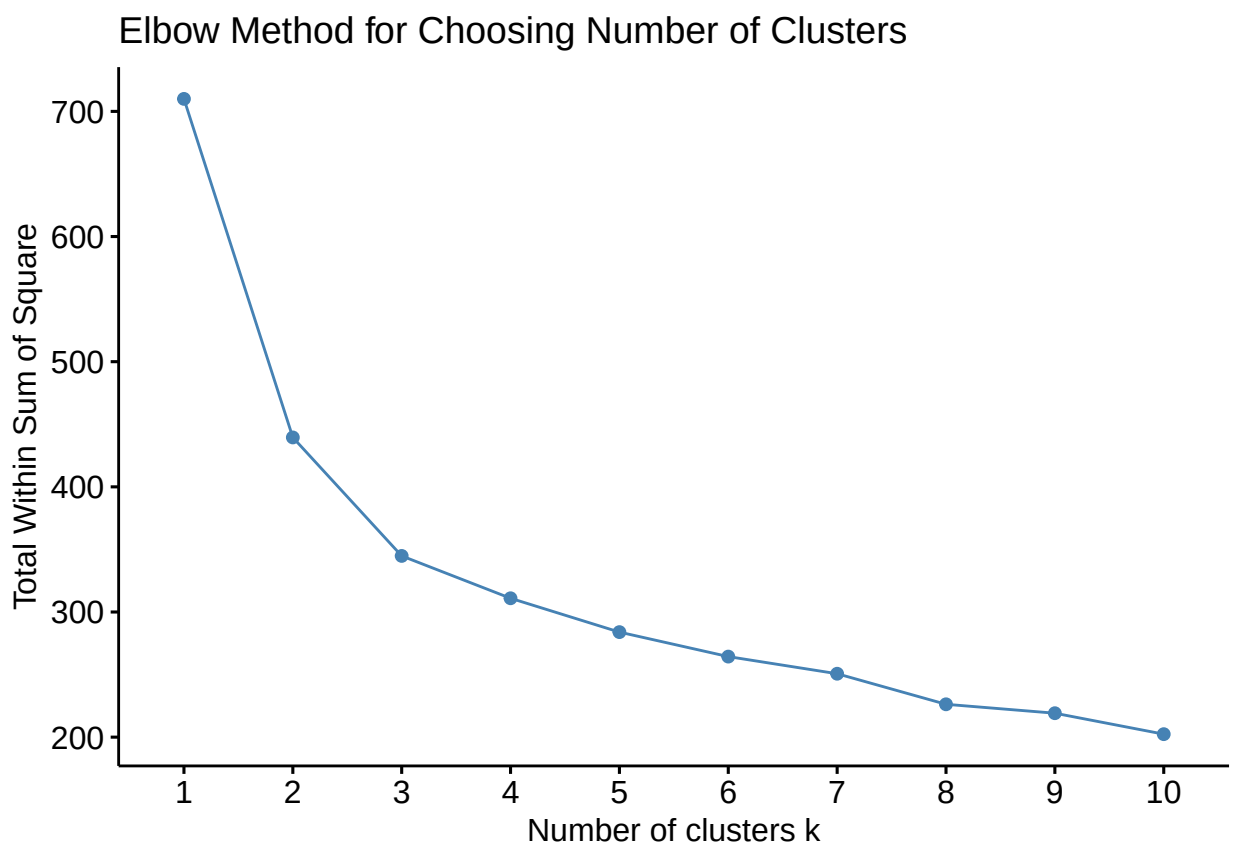
The correlation heatmap shows that many workplace culture variables are moderately to strongly positively correlated. In particular, culture, leadership, and overall ratings exhibit strong relationships, suggesting that they capture a shared underlying dimension of workplace quality.

In contrast, variables such as business outlook and CEO approval are less strongly correlated with the core culture metrics, indicating that they may capture distinct aspects of employee perception.

This pattern helps explain why clustering produces a strong primary separation, as many features move together, while still allowing for meaningful variation across companies.

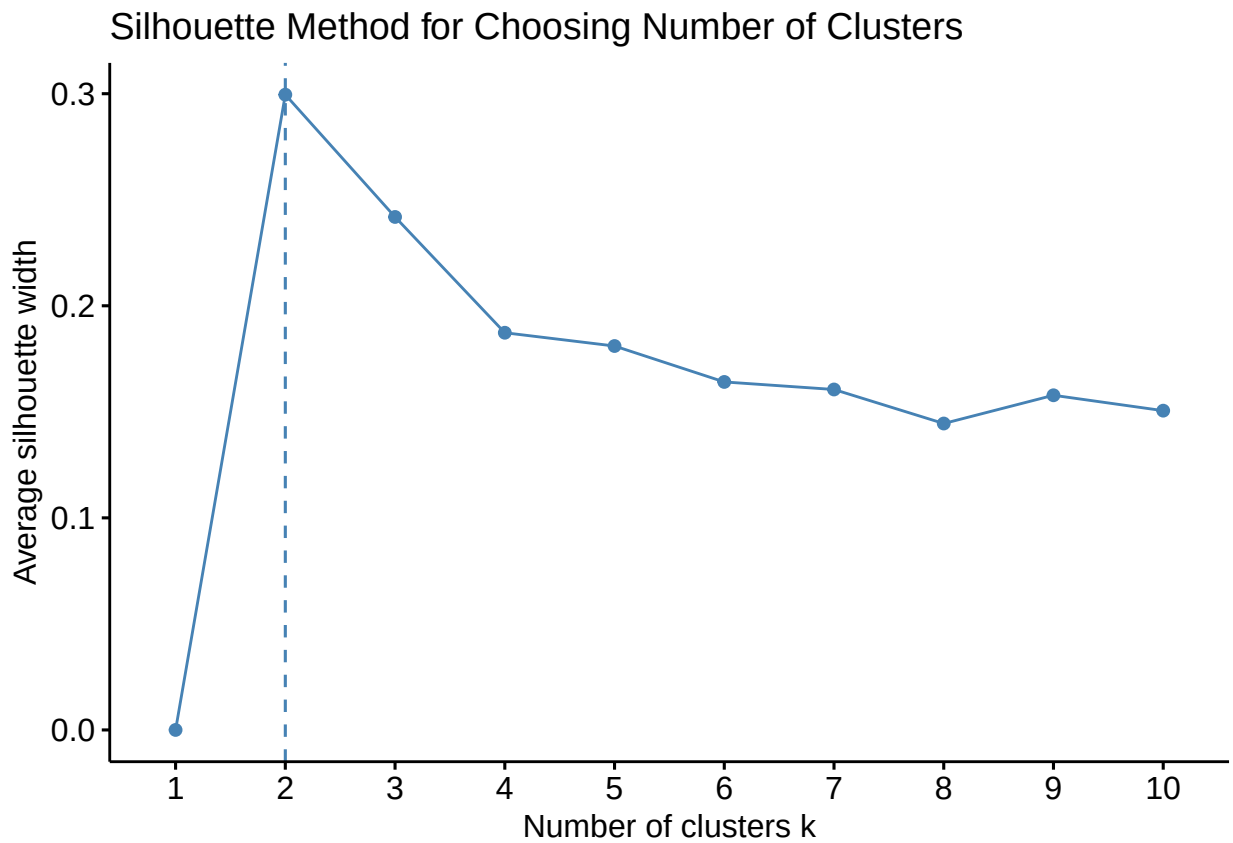
Elbow Method

```
set.seed(123)

fviz_nbclust(scaled_features, kmeans, method = "wss") +
  labs(title = "Elbow Method for Choosing Number of Clusters")
```



Silhouette Score

```
fviz_nbclust(scaled_features, kmeans, method = "silhouette") +
  labs(title = "Silhouette Method for Choosing Number of Clusters")
```



Compare multiple k values

```
set.seed(123)

k_grid <- 2:4

k_results <- purrr::map_dfr(k_grid, function(k) {
  km <- kmeans(scaled_features, centers = k, nstart = 25)
  sil_obj <- silhouette(km$cluster, dist(scaled_features))

  tibble(
    k = k,
    tot_withinss = km$tot.withinss,
    betweenss_ratio = km$betweenss / km$totss,
    avg_silhouette = mean(sil_obj[, 3])
  )
})

k_results
```

```
## # A tibble: 3 x 4
##       k tot_withinss betweenss_ratio avg_silhouette
##   <int>        <dbl>           <dbl>          <dbl>
## 1     2         433.           0.391          0.300
```

```
## 2     3         344.           0.515          0.242
## 3     4         306.           0.569          0.187
```

A comparison of clustering solutions for k = 2, 3, and 4 shows that the two-cluster solution achieves the highest silhouette score, indicating the strongest separation between groups. However, it explains less variance in the data, as reflected by the lower between-cluster sum of squares ratio.

The three-cluster solution provides a balance between cluster separation and explanatory power. While its silhouette score is slightly lower, it captures a larger proportion of the variance and produces a more interpretable segmentation into low, moderate, and high workplace culture groups.

The four-cluster solution further increases explained variance but results in a noticeable drop in silhouette score, suggesting over-segmentation and less stable clusters.

Based on this trade-off, the three-cluster solution was selected as the final model.

2. K-means Model

```r
set.seed(123)

kmeans_model <- kmeans(scaled_features, centers = 3, nstart = 25)

company_data_filtered$cluster <- factor(kmeans_model$cluster)

# View cluster sizes
table(company_data_filtered$cluster)
```

```
##
##  1  2  3
## 14 38 20
```

Cluster quality metrics

```r
# Silhouette scores
sil <- silhouette(kmeans_model$cluster, dist(scaled_features))

# Average silhouette width
avg_sil_width <- mean(sil[, 3])
avg_sil_width
```

```
## [1] 0.2418881
```

```r
# Between-cluster sum of squares ratio
between_total_ratio <- kmeans_model$betweenss / kmeans_model$totss
between_total_ratio
```

```
## [1] 0.5154375
```

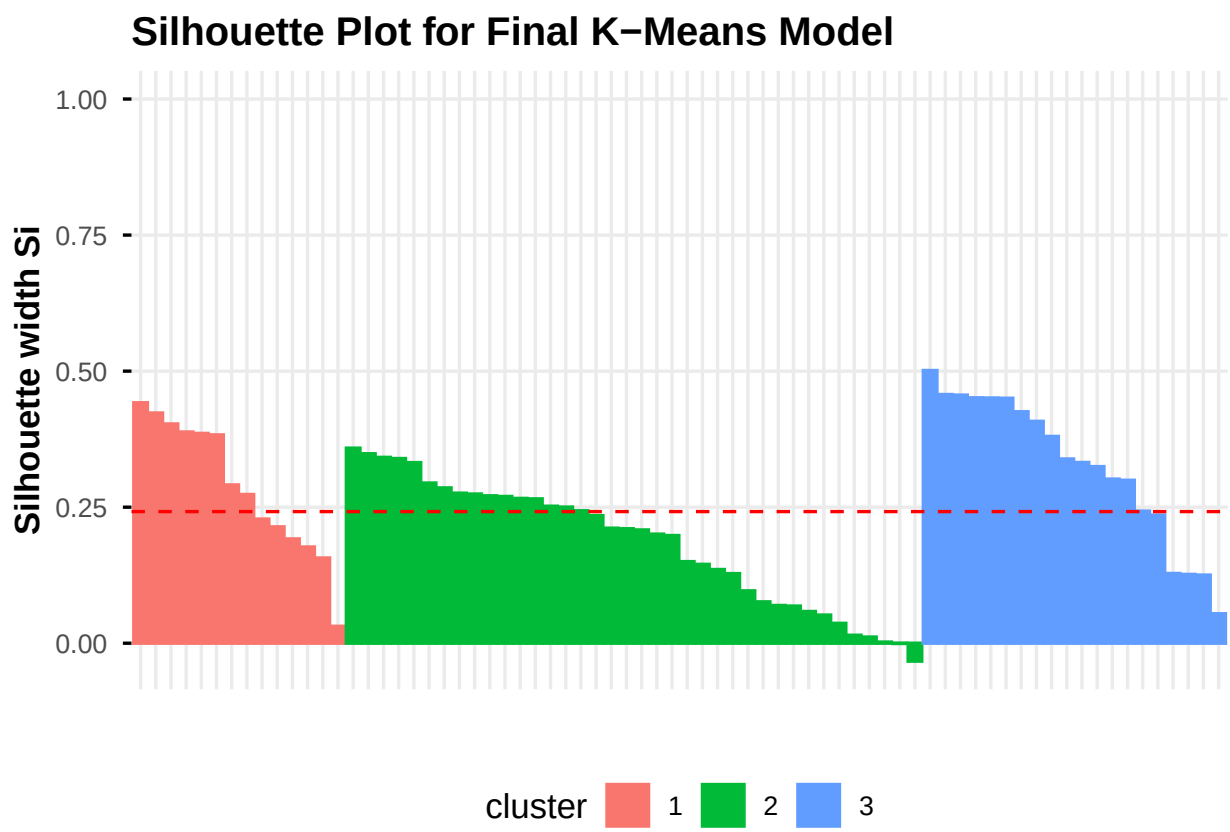
Overall, the clustering results indicate moderate structure in the data. While the silhouette score suggests that cluster separation is not very strong, the between-to-total sum of squares ratio shows that a substantial portion of the variation is captured by the clustering. This implies that the identified clusters provide a useful, though not perfectly distinct, grouping of companies based on workplace culture.

The relatively low silhouette score reflects the continuous nature of workplace culture, where companies may share overlapping characteristics rather than forming sharply separated groups.

Silhouette plot for the final model

```
fviz_silhouette(sil) +
  labs(title = "Silhouette Plot for Final K-Means Model")
```

```
##   cluster size ave.sil.width
## 1       1   14          0.28
## 2       2   38          0.18
## 3       3   20          0.32
```



The silhouette plot provides a company-level view of cluster membership quality. Companies with larger silhouette values are well matched to their assigned cluster, while companies with values near zero lie closer to the boundary between clusters.

Looking at Cluster Centers

```r
cluster_summary <- company_data_filtered %>%
  group_by(cluster) %>%
  summarise(
    work_life = mean(work_life),
    culture = mean(culture),
    leadership = mean(leadership),
    career = mean(career),
    compensation = mean(compensation),
    overall = mean(overall),
    diversity = mean(diversity),
    recommend = mean(recommend),
    outlook = mean(outlook),
    ceo_approval = mean(ceo_approval),
    companies_in_cluster = n()
  )

cluster_summary_round <- cluster_summary %>%
  mutate(across(where(is.numeric), ~ round(., 2)))

cluster_summary_round
```

```
## # A tibble: 3 x 12
##   cluster work_life culture leadership career compensation overall diversity
##   <fct>       <dbl>   <dbl>      <dbl>  <dbl>        <dbl>   <dbl>     <dbl>
## 1 1            1.78    1.82       1.58   2.34         2.18    1.79      2.37
## 2 2            2.97    2.73       2.41   2.99         3.23    2.87      3.56
## 3 3            4.12    3.91       3.62   3.7          3.95    3.91      4.13
## # i 4 more variables: recommend <dbl>, outlook <dbl>, ceo_approval <dbl>,
## #   companies_in_cluster <dbl>
```

The three clusters can be interpreted as broad workplace culture groups.

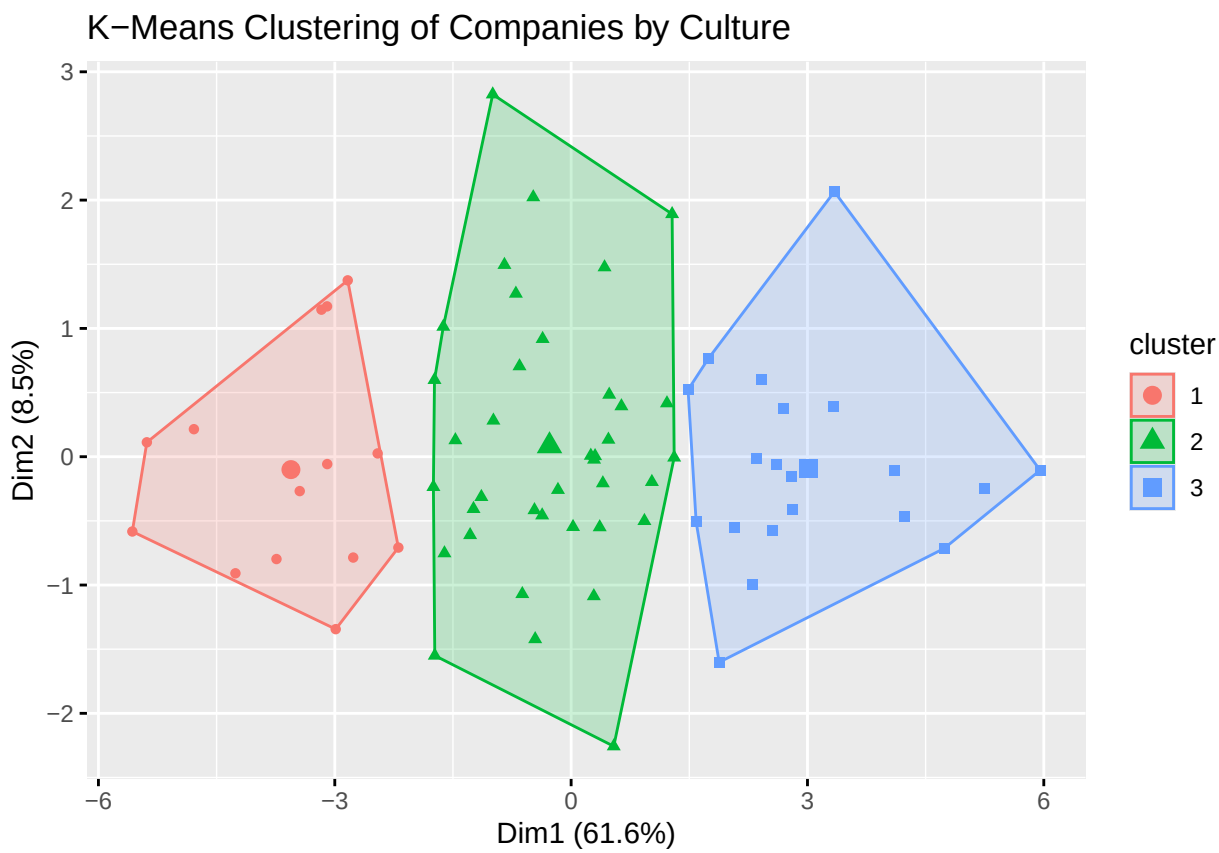
Cluster 1 represents low-rated companies. These companies have the weakest scores across work-life balance, culture and values, leadership, and overall satisfaction. This cluster reflects less favorable employee experiences.

Cluster 2 represents moderate companies. They have mid-range scores across most culture-related dimensions. They appear to provide average employee experiences, with neither especially poor nor especially strong workplace culture outcomes.

Cluster 3 represents high-rated companies. These companies score highest on work-life balance, culture and values, leadership, compensation, and overall satisfaction. This cluster captures firms with the most favorable employee-reported culture profiles.

Visualize the Clusters

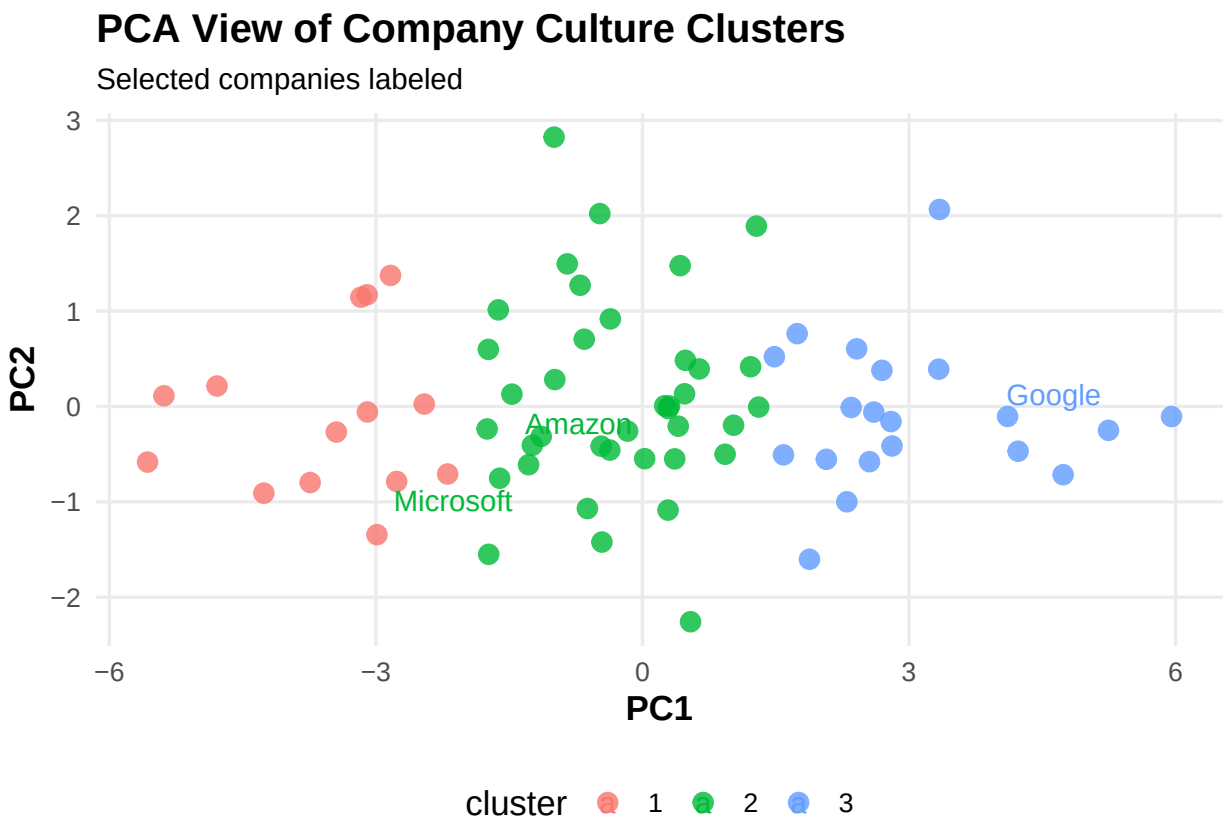
```r
fviz_cluster(kmeans_model, data = scaled_features,
             geom = "point",
             ellipse.type = "convex",
             main = "K-Means Clustering of Companies by Culture")
```



PCA Plot

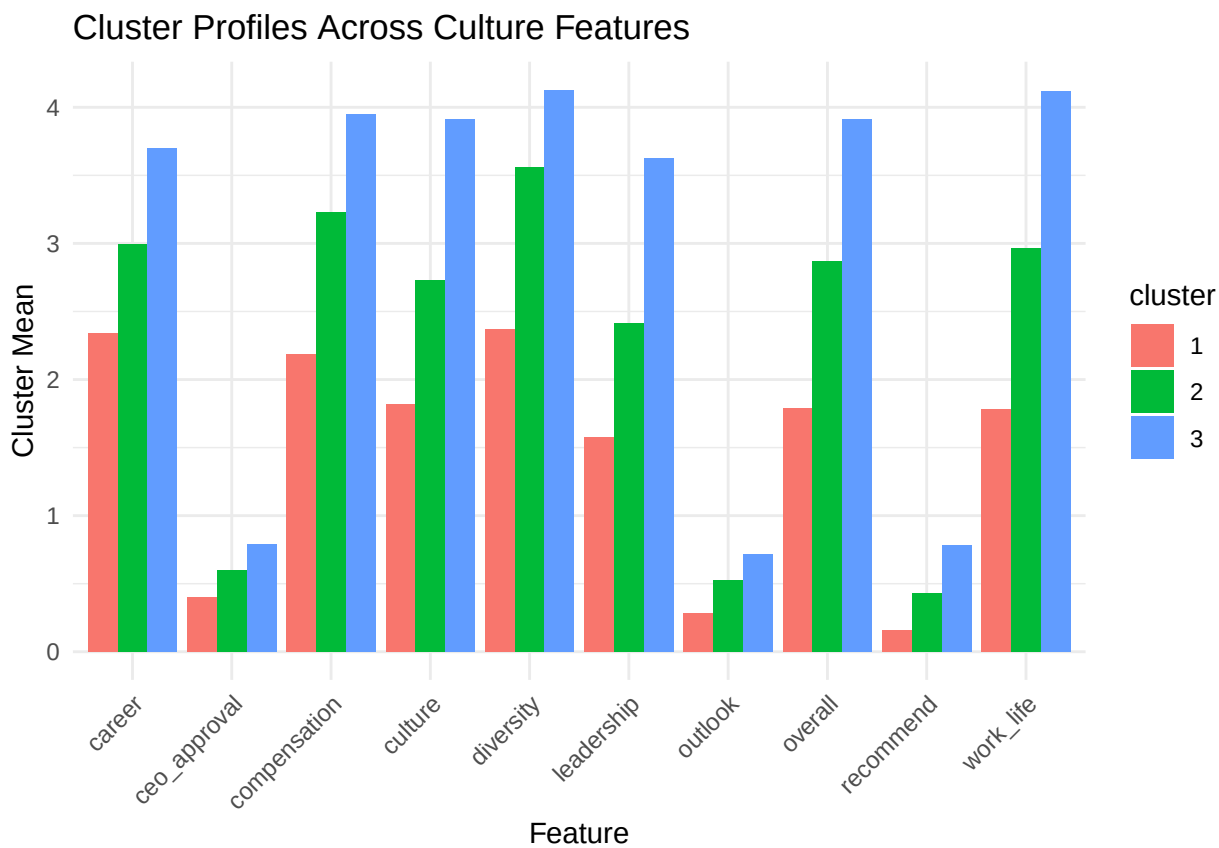
```
pca_df <- prcomp(scaled_features)

pca_scores <- as.data.frame(pca_df$x[, 1:2])
pca_scores$company_name <- company_data_filtered$company_name
pca_scores$cluster <- company_data_filtered$cluster

highlight_companies <- c("Google", "Microsoft", "Apple", "Amazon", "Meta")

pca_scores$label <- ifelse(
  pca_scores$company_name %in% highlight_companies,
  pca_scores$company_name,
  NA
)

ggplot(pca_scores, aes(x = PC1, y = PC2, color = cluster)) +
  geom_point(size = 3, alpha = 0.8) +
  geom_text_repel(aes(label = label), na.rm = TRUE) +
  labs(
    title = "PCA View of Company Culture Clusters",
    subtitle = "Selected companies labeled",
    x = "PC1",
    y = "PC2"
```

```
  )
```



Cluster Profile Plot

```
cluster_profile_long <- cluster_summary %>%
  select(cluster, work_life, culture, leadership, career,
         compensation, overall, diversity, recommend, outlook, ceo_approval) %>%
  pivot_longer(
    cols = -cluster,
    names_to = "feature",
    values_to = "mean_value"
  )

ggplot(cluster_profile_long, aes(x = feature, y = mean_value, fill = cluster)) +
  geom_col(position = "dodge") +
  theme_minimal() +
  labs(
    title = "Cluster Profiles Across Culture Features",
    x = "Feature",
    y = "Cluster Mean"
  ) +
  theme(axis.text.x = element_text(angle = 45, hjust = 1))
```



Find companies similar to a chosen company

```
target_company <- "Google"

target_cluster <- company_data_filtered %>%
  filter(company_name == target_company) %>%
  pull(cluster)

similar_companies <- company_data_filtered %>%
  filter(cluster == target_cluster) %>%
  arrange(desc(work_life), desc(culture))

similar_companies
```

```
## # A tibble: 20 x 13
##    company_name        work_life culture leadership career compensation overall
##    <chr>                   <dbl>   <dbl>      <dbl>  <dbl>        <dbl>   <dbl>
##  1 American Express         5       5          4.5    3.5          4       4.5
##  2 Johnson & Johnson        5       4.67       5      4            5       4.67
##  3 Airbus                   5       4          4      3.5          5       4.5
##  4 Merck                    4.5     4.5        3      3            4       4
##  5 UBS                      4.5     4.33       4.17   3.67         3.67    3.5
##  6 Sony                     4.5     4          4.5    4            4       4
##  7 Capgemini                4.5     3.5        3.5    3            3       3.5
```

```
##  8 Caterpillar              4.33    3.67       3      4.33         4       4
##  9 Stellantis               4       4.5        4      3.5          4.5     4
## 10 Morgan Stanley           4       4.33       3.67   3.67         3.67    3.67
## 11 Google                   4       4.25       3.88   4.25         4.38    4.12
## 12 Southwest Airlines       4       4          3.67   3.67         4       3.67
## 13 Novartis                 4       3.5        2.67   3            4.17    3.17
## 14 RBC                      4       3.5        3.5    4            3.75    3.75
## 15 BBVA                     4       3          2.5    4            4       4
## 16 Ford Motor Company       3.67    3.67       3.67   4.33         4.33    4
## 17 General Motors (GM)      3.5     3.75       3.75   4            4       4.25
## 18 Accenture                3.45    3.73       3.36   3.64         3.36    3.55
## 19 JPMorgan Chase & Co      3.44    3.33       3.17   3.39         3.67    3.44
## 20 HDFC Bank                3       3          3      3.5          2.5     4
## # i 6 more variables: diversity <dbl>, recommend <dbl>, outlook <dbl>,
## #   ceo_approval <dbl>, review_count <int>, cluster <fct>
```

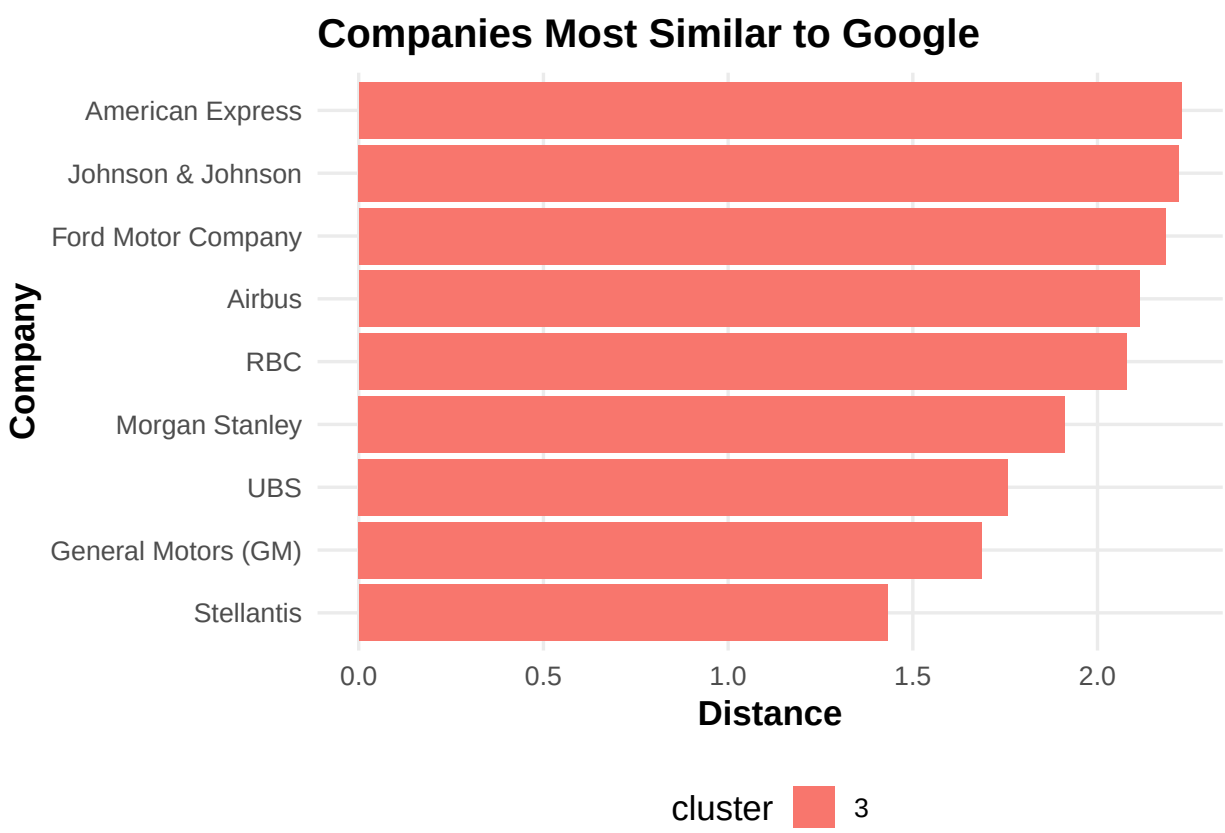
Find the most similar companies by distance

```r
scaled_df <- as.data.frame(scaled_features)
scaled_df$company_name <- company_data_filtered$company_name

target_company <- "Google"

target_row <- scaled_df %>%
  filter(company_name == target_company) %>%
  select(-company_name)

nearest_companies <- scaled_df %>%
  rowwise() %>%
  mutate(distance = sqrt(sum((c_across(-company_name) - as.numeric(target_row))^2))) %>%
  ungroup() %>%
  left_join(
    company_data_filtered %>%
      select(company_name, cluster, overall, culture, work_life, review_count),
    by = "company_name"
  ) %>%
  arrange(distance) %>%
  slice(1:10)

nearest_companies
```

```
## # A tibble: 10 x 17
##    work_life.x culture.x leadership career compensation overall.x diversity
##          <dbl>     <dbl>      <dbl>  <dbl>        <dbl>     <dbl>     <dbl>
##  1       0.956     1.52        1.42  1.81         1.33      1.36      1.11
##  2       0.956     1.79        1.56  0.668        1.48      1.22      1.48
##  3       0.450     0.963       1.28  1.43         0.897     1.51      0.258
##  4       1.46      1.61        1.74  0.922        0.512     0.639     1.15
##  5       0.956     1.61        1.19  0.922        0.512     0.832     0.502
##  6       0.956     0.687       1.01  1.43         0.608     0.929     0.990
##  7       1.97      1.24        1.56  0.668        2.06      1.80      0.502
##  8       0.618     0.871       1.19  1.94         1.28      1.22      0.990
```

```
## 9        1.97      1.98        2.66  1.43         2.06      1.99      1.48
## 10       1.97      2.34        2.11  0.668        0.897     1.80      1.48
## # i 10 more variables: recommend <dbl>, outlook <dbl>, ceo_approval <dbl>,
## #   company_name <chr>, distance <dbl>, cluster <fct>, overall.y <dbl>,
## #   culture.y <dbl>, work_life.y <dbl>, review_count <int>
```

```
ggplot(nearest_companies %>% filter(company_name != target_company),
       aes(x = reorder(company_name, distance), y = distance, fill = cluster)) +
  geom_col() +
  coord_flip() +
  labs(
    title = "Companies Most Similar to Google",
    x = "Company",
    y = "Distance"
  )
```



Feature Contribution to Similarity

```
# Compute per-feature squared differences
feature_contributions <- scaled_df %>%
  filter(company_name != target_company) %>%
  mutate(across(-company_name,
                ~ (. - as.numeric(target_row[[cur_column()]]))^2)) %>%
  mutate(total_distance = rowSums(across(-company_name))) %>%
```

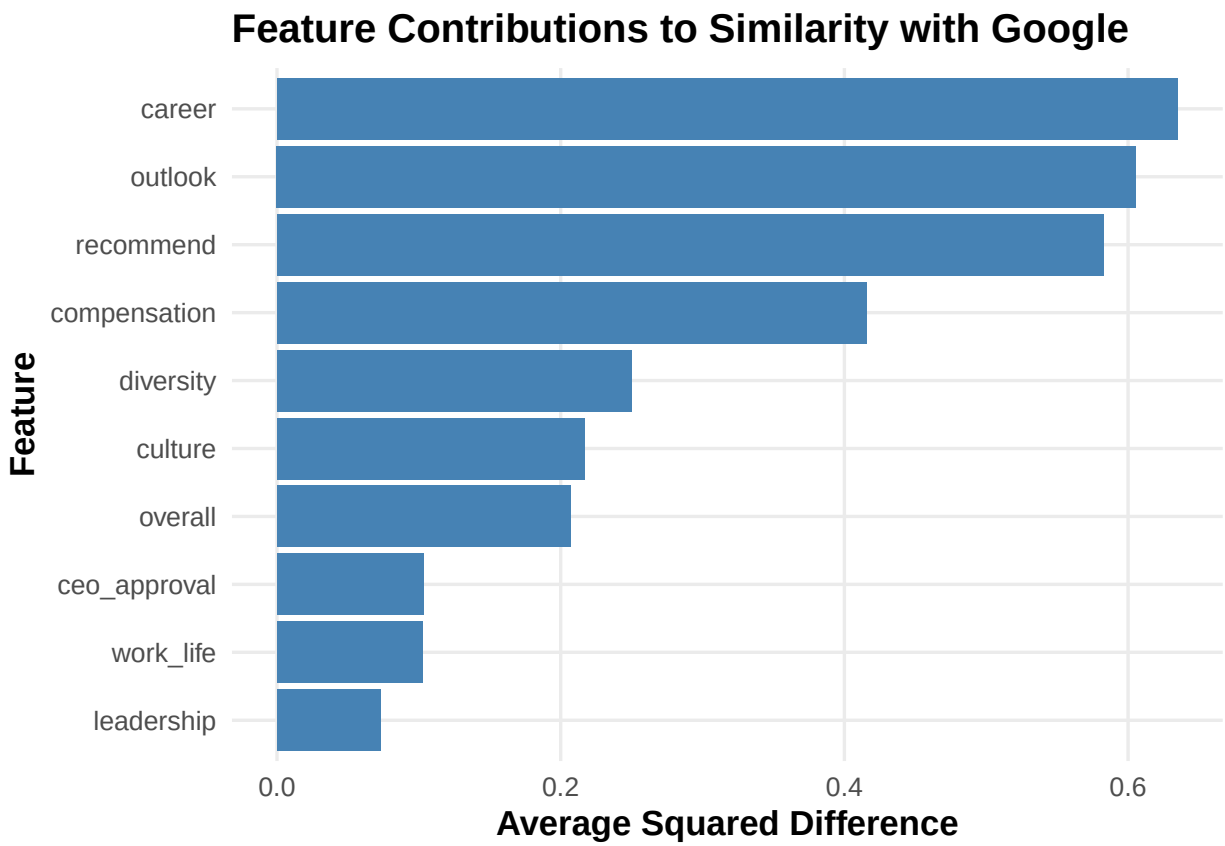
```r
  arrange(total_distance)

# Top 5 most similar companies
top_similar <- feature_contributions %>%
  slice(1:5)

top_similar
```

```
##   work_life     culture leadership    career compensation    overall
## 1 0.0000000 0.076320997 0.01897532 1.3054018   0.02094497 0.02094249
## 2 0.2567037 0.305283986 0.01897532 0.1450446   0.18850472 0.02094249
## 3 0.2567037 0.008480111 0.10331010 0.7896875   0.67256622 0.52356226
## 4 0.0000000 0.008480111 0.05270923 0.7896875   0.67256622 0.28156015
## 5 0.0000000 0.686888969 0.17077791 0.1450446   0.52362422 0.18848241
##     diversity recommend   outlook ceo_approval        company_name
## 1 0.133886026 0.1759083 0.2369697  0.060549180          Stellantis
## 2 0.728935028 0.1759083 0.9478788  0.060549180 General Motors (GM)
## 3 0.001652914 0.4886342 0.2369697  0.006727687                 UBS
## 4 0.371905627 0.4886342 0.6582492  0.329656649      Morgan Stanley
## 5 0.014876225 1.5831748 0.9478788  0.060549180                 RBC
##   total_distance
## 1       2.049899
## 2       2.848726
## 3       3.088294
## 4       3.653449
## 5       4.321297
```

```r
# Average contribution per feature
feature_importance <- top_similar %>%
  select(-company_name, -total_distance) %>%
  summarise(across(everything(), mean)) %>%
  pivot_longer(cols = everything(),
               names_to = "feature",
               values_to = "avg_contribution") %>%
  arrange(avg_contribution)

feature_importance
```

```
## # A tibble: 10 x 2
##    feature      avg_contribution
##    <chr>                   <dbl>
##  1 leadership             0.0729
##  2 work_life              0.103
##  3 ceo_approval           0.104
##  4 overall                0.207
##  5 culture                0.217
##  6 diversity              0.250
##  7 compensation           0.416
##  8 recommend              0.582
##  9 outlook                0.606
## 10 career                 0.635
```

```r
ggplot(feature_importance,
       aes(x = reorder(feature, avg_contribution),
           y = avg_contribution)) +
  geom_col(fill = "steelblue") +
  coord_flip() +
  labs(
    title = "Feature Contributions to Similarity with Google",
    x = "Feature",
    y = "Average Squared Difference"
  )
```



To better understand what drives similarity, the Euclidean distance was decomposed into feature-level contributions, allowing identification of which workplace culture dimensions are most similar between Google and its nearest companies.

Limitations of K-Means

This analysis has several limitations.

First, some companies have only a small number of reviews, which may make their average culture ratings less stable. Although a minimum threshold of two reviews was used to preserve sample size, this still introduces some uncertainty.

Second, all selected features were treated as equally important in the clustering process after scaling, even though work-life balance and culture values may be more central to the research question than other variables.

Third, K-means clustering requires the number of clusters to be chosen in advance, so the final solution depends partly on the selected value of k.

Finally, the results reflect only the available Glassdoor reviews in this sample and may not fully capture the broader culture of each company.

Conclusion

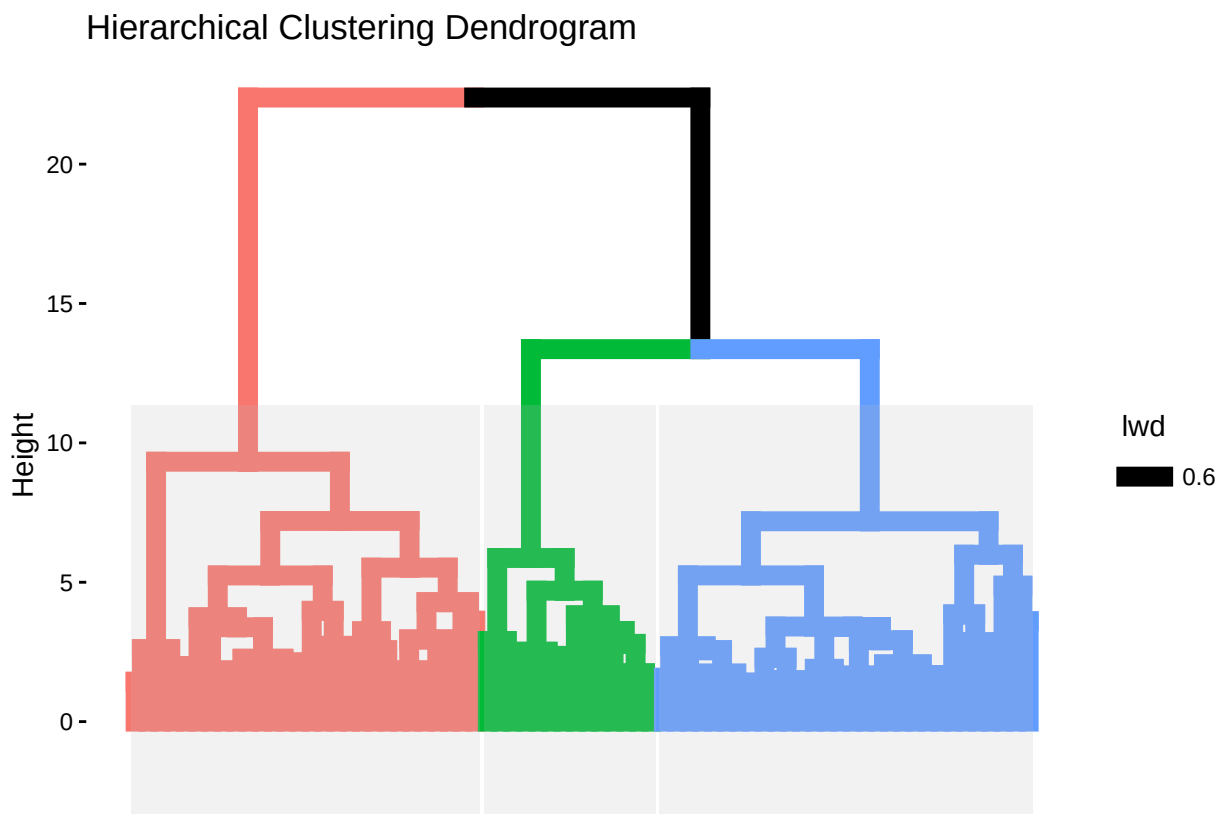
Overall, the K-means clustering analysis identified three broad groups of companies based on employee-reported workplace culture characteristics. The results suggest that companies can be grouped into low, moderate, and high-culture profiles using features such as work-life balance, culture and values, leadership, compensation, and recommendation rates. By combining cluster membership with distance-based similarity, the analysis provides a pragmatic way to identify companies with workplace cultures similar to a target firm.

3. Hierarchical Clustering

```r
dist_matrix <- dist(scaled_features, method = "euclidean")

hc_model <- hclust(dist_matrix, method = "ward.D2")
```

Dendrogram

```r
fviz_dend(
  hc_model,
  k = 3,
  rect = TRUE,
  rect_fill = TRUE,
  show_labels = FALSE,
  lwd = 0.6,
  main = "Hierarchical Clustering Dendrogram"
)
```



The dendrogram shows a large separation at higher levels of the hierarchy, indicating that some groups of companies are more distinctly separated, while finer structure exists within these broader groups.

Cut into clusters

```
hc_clusters <- cutree(hc_model, k = 3)

company_data_filtered$hc_cluster <- factor(hc_clusters)

# Compare with K-means
table(KMeans = company_data_filtered$cluster,
      Hierarchical = company_data_filtered$hc_cluster)
```

```
##       Hierarchical
## KMeans  1  2  3
##      1  1  0 13
##      2 29  8  1
##      3  0 20  0
```

```
sil_hc <- silhouette(hc_clusters, dist(scaled_features))
data.frame(
  sil_h = mean(sil_hc[,3]),
  sil_k = mean(sil[,3])
)
```

```
##       sil_h     sil_k
## 1 0.2085152 0.2418881
```

To compare clustering quality across methods, silhouette scores were computed for both K-means and hierarchical clustering. The K-means model achieved a slightly higher silhouette score (0.242) compared to hierarchical clustering (0.209), indicating that it provides better separation between clusters.

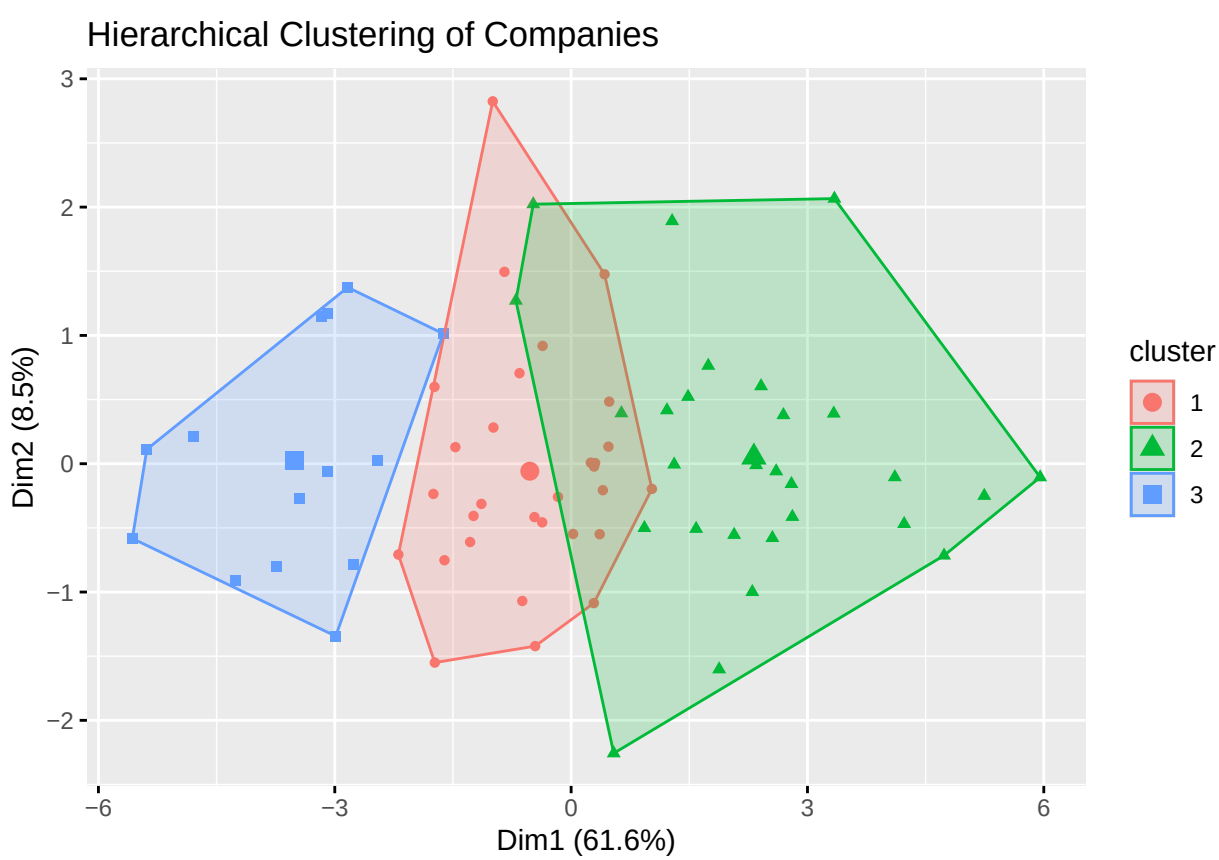
A comparison between K-means and hierarchical clustering shows that the cluster assignments are largely consistent across methods. In particular, one cluster is perfectly aligned between the two approaches, indicating a stable group of companies with similar culture characteristics.

However, some variation exists in the middle cluster, where companies are distributed across multiple hierarchical groups. This suggests that companies with moderate workplace culture characteristics are less distinctly separated, reflecting the continuous nature of culture-related variables.

Overall, the general agreement between the two clustering methods supports the robustness of the identified company groupings.

Hierarchical cluster scatterplot

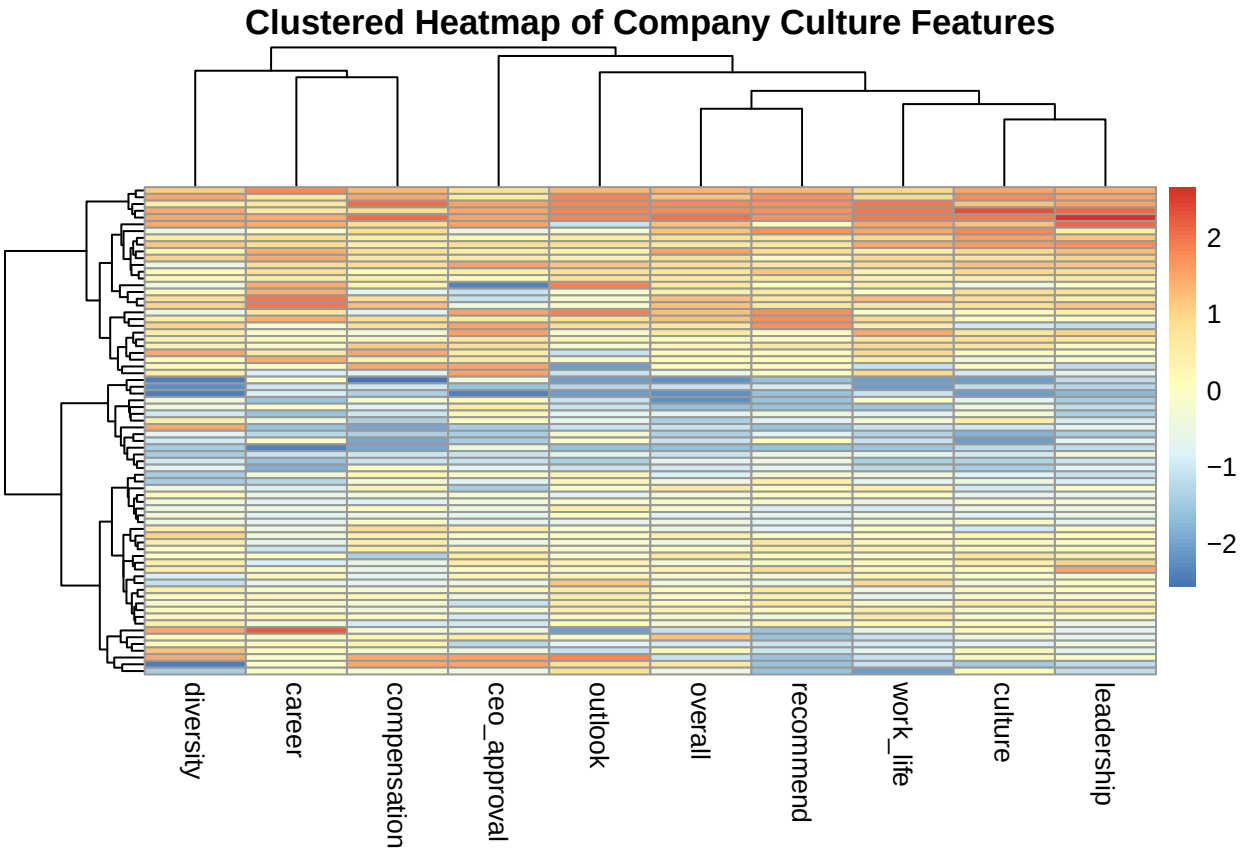
```r
fviz_cluster(
  list(data = scaled_features, cluster = hc_clusters),
  geom = "point",
  ellipse.type = "convex",
  main = "Hierarchical Clustering of Companies"
)
```



The hierarchical cluster visualization shows clear separation along the first principal dimension, which explains the majority of the variation in the data (61.6%). Companies in the high-culture cluster are well separated from the rest, while moderate companies show some overlap with adjacent groups. This pattern is consistent with earlier findings that workplace culture varies along a continuum rather than forming perfectly distinct categories.

Heatmap of clustered companies

```
pheatmap(
  scale(clustering_features),
  cluster_rows = hc_model,
  cluster_cols = TRUE,
  show_rownames = FALSE,
  main = "Clustered Heatmap of Company Culture Features"
)
```



The heatmap groups companies based on similarity rather than ordering them from highest to lowest. One group of companies shows consistently below-average values across most features (blue tones), representing lower workplace culture ratings. Another group shows values closer to the average (white tones), indicating moderate workplace culture. A third group exhibits higher values across features (warmer tones), representing stronger workplace culture profiles.

This confirms that the clustering captures meaningful differences in company culture, even though the ordering of groups in the heatmap is not based on ranking.

Limitations of Hierarchical Clustering

Hierarchical clustering also has several limitations. First, the method is sensitive to the choice of distance metric and linkage criteria, meaning that different specifications (e.g., Euclidean distance or alternative linkage methods) may produce different cluster structures.

Second, hierarchical clustering is based on a sequence of irreversible merges, so early grouping decisions cannot be adjusted later in the process. This can lead to suboptimal cluster assignments if initial merges are not ideal.

Additionally, the method can be sensitive to outliers, which may distort the dendrogram structure and influence cluster formation. Finally, unlike K-means, hierarchical clustering does not provide a clear objective criterion for selecting the number of clusters, requiring subjective interpretation of the dendrogram.